Name: _____

Class: _____

Date: _____

Mr. Rawson • GCSE Physics

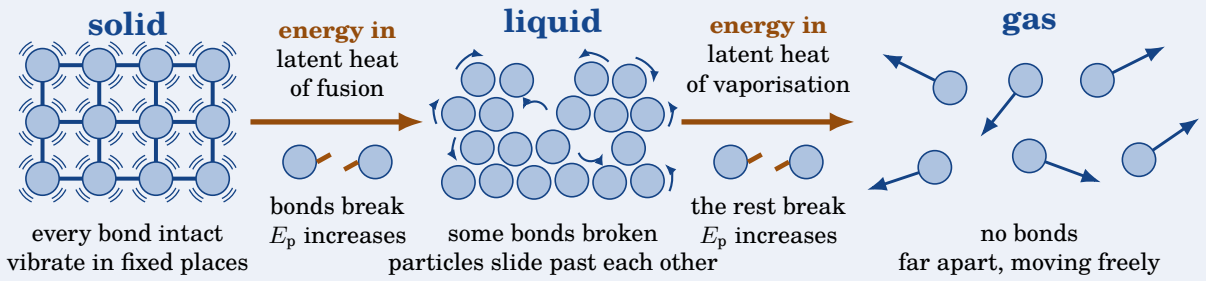
3.5 Specific Latent Heat

1 · Latent heat energy in, temperature constant

Key concept — latent heat

Latent heat is the energy needed for a substance to change state. While it melts, boils, freezes or condenses, its temperature stays constant. On your Heating Curve: (B)(D) the flat sections.

The energy breaks bonds between the particles. Pulling attracting particles apart stores energy, as lifting a book does: their E_p increases. Their E_k stays the same, so the temperature stays the same.



1) **Explain** why the temperature of a block of ice stays at $0\,°C$ while it melts, even though it is being heated the whole time. Use the words *bonds*, *potential energy* and *kinetic energy*. **[3]**

The energy supplied is used to weaken / break the bonds between the particles, so the potential energy of the particles increases. Their kinetic energy does not change, so they move no faster, and the temperature (which depends on the kinetic energy) stays the same.

2 · Specific latent heat $E = m\,L$

Key concept — specific latent heat

Specific latent heat, L, is the energy needed to change the state of one kilogram of a substance with no change in temperature.

Specific latent heat of fusion is for melting: solid → liquid (and freezing).

Specific latent heat of vaporisation is for boiling: liquid → vapour (and condensing).

$$E = m\,L$$

- E: **thermal energy for a change of state** (J)
- m: **mass** (kg)
- L: **specific latent heat** (J/kg)

(B) (D)

Specific latent heat

Substance	Fusion J/kg	Vaporisation J/kg
Water	334 000	2 260 000
Ethanol	108 000	840 000
Lead	23 000	870 000
Aluminium	400 000	10 500 000

No need to memorise these. In an exam, the question gives you the value you need.

You will use them in calculations: use these values in every question on this sheet.

Worked example 1 — find E

Calculate the energy needed to melt 0.25 kg of ice at 0 °C, turning it into 0.25 kg of liquid water at 0 °C.

1. Which L? Solid → liquid, so fusion. Water: 334 000 J/kg

2. Equation $E = m\,L$

3. Substitute $E = 0.25 \times 334\,000$

4. Answer $E =$ **83 500 J**

Worked example 2 — find m

A kettle transfers 452 000 J to water at 100 °C. Calculate the mass of water that boils away, turning from liquid water into steam (gas).

1. Which L? Boiling, so vaporisation. Water: 2 260 000 J/kg

2. Rearrange divide by L: $m = \dfrac{E}{L}$

3. Substitute $m = \dfrac{452\,000}{2\,260\,000}$

4. Answer $m =$ **0.20 kg**

2) **Calculate** the energy needed to boil 2.0 kg of water that is already at 100 °C, turning 2.0 kg of liquid water into 2.0 kg of steam (gas). **[2]**

Liquid → vapour, so vaporisation: $E = m\,L = 2.0 \times 2\,260\,000 =$ **4 520 000 J** *(4.52 MJ).*

3) If one were to apply 167 000 J of energy to a block of ice at 0 °C, how much ice (in kg) would melt from solid to liquid? **[2]**

$m = \dfrac{E}{L} = \dfrac{167\,000}{334\,000} =$ **0.50 kg**

Worked example 3 — find L

It takes 120 000 J to melt 0.30 kg of a metal at its melting point, turning it from solid to liquid. Calculate the specific latent heat of fusion of the metal.

1. Rearrange divide by m: $L = \dfrac{E}{m}$

2. Substitute $L = \dfrac{120\,000}{0.30}$

3. Answer $L =$ **400 000 J/kg**: the table says aluminium.

Worked example 4 — mass in grams

Calculate the energy needed to melt 50 g of ice at 0 °C, turning it into 50 g of liquid water.

1. Convert $50\text{ g} = 50 \div 1000 = 0.050\text{ kg}$

2. Equation $E = m\,L$

3. Substitute $E = 0.050 \times 334\,000$

4. Answer $E =$ **16 700 J**

4) **Calculate** the specific latent heat of vaporisation of a liquid, given that 42 000 J boils away 0.050 kg of it, turning 0.050 kg of liquid into 0.050 kg of gas. Use the table to identify the liquid. **[3]**

$L = \dfrac{E}{m} = \dfrac{42\,000}{0.050} =$ **840 000 J/kg** — *ethanol.*

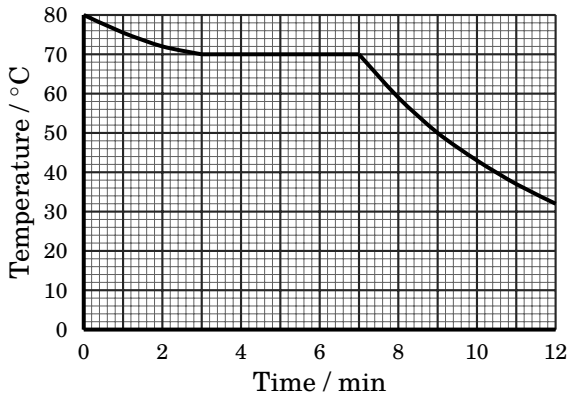
5) **Calculate** the energy needed to boil away 250 g of water at 100 °C, turning 250 g of liquid water into 250 g of steam (gas). **[3]**

$250\,g = 0.25\,kg$. $E = m\,L = 0.25 \times 2\,260\,000 =$ **565 000 J**

Check every answer in this section: mass in **kg**, a **unit** on the answer, and boiling needs several times more energy than melting the same mass.

3 · Cooling graphs — your Heating Curve in reverse

Cooling is your Heating Curve in reverse. Graph 1 shows liquid stearic acid (a wax) left to cool in a test tube standing in a beaker of cold water.



Graph 1 — stearic acid cooling.

6) **Write down** the melting point of stearic acid. **[1]**

70 °C — *the temperature of the flat section.*

7) **Write down** the time at which the wax became completely solid. **[1]**

7 minutes — the end of the flat section.

8) **Explain** why the temperature stays constant between 3 and 7 minutes, even though the wax is transferring energy to the water around it the whole time. **[2]**

The wax is freezing. As bonds form between the particles their potential energy falls, and that is the energy transferred to the surroundings. The kinetic energy of the particles stays the same, so the temperature stays the same.

4 · Multi-phase questions — more than one stage on the Heating Curve

Higher *a two-stage problem*

9) Someone puts 5 ice cubes in a glass, forgets about it, and leaves it for 12 hours. The ice absorbs energy from the room around it. First it melts, then the water warms up until it is in thermal equilibrium with the room: the same temperature, 20 °C.

Each ice cube has a mass of 0.10 kg and starts at 0 °C. How much energy has **all** of the ice absorbed, to turn it from solid ice at 0 °C into liquid water at 20 °C? **[5]**

Stage 1 — melt the ice
(change of state, $E = m\,L$)

$m = 5 \times 0.10 = 0.50\,\text{kg}$
$E_1 = m\,L = 0.50 \times 334\,000$
$E_1 = \mathbf{167\,000\ J}$

Stage 2 — warm the water
(temperature rise, $\Delta E = m\,c\,\Delta\theta$)

$E_2 = m\,c\,\Delta\theta$
$E_2 = 0.50 \times 4200 \times 20$
$E_2 = \mathbf{42\,000\ J}$

Total energy $= E_1 + E_2 =$ _____ $167\,000 + 42\,000 = \mathbf{209\,000\ J}$ *(209 kJ). 1 mark each: E_1 substitution, E_1, E_2 substitution, E_2, total with unit.*

5 · Energy from motion and height — macro energy becomes micro energy

Higher *one energy chain, written as one equation*

A moving or falling object that stops can turn its E_k or E_p (macro) into internal energy (micro). Write the chain as **one equation**, e.g. $\frac{1}{2}\,m\,v^2 = m\,L$: the whole object's mass on the left, the mass that melts on the right.

10) **Calculate** the mass of snow that melts when a 280 g snowball at 0 °C hits a wall at 25 m/s and stops. Assume all of its kinetic energy is used to melt snow. Give your answer in grams. **[4]**

$E_k = \frac{1}{2}\,m\,v^2 = \frac{1}{2} \times 0.280 \times 25^2 = 87.5\,J.\ \ m = \frac{E}{L} = \frac{87.5}{334\,000} = 0.000\,26\,kg = \mathbf{0.26\ g}.$

11) **Calculate** the percentage of a 4.0 g hailstone at 0 °C that melts when it falls 2500 m. Assume all of its gravitational potential energy is used to melt it. $g = 9.8$ N/kg. **[5]**

$E_p = m\,g\,h = 0.0040 \times 9.8 \times 2500 = 98\,J.\ \ m = \frac{E}{L} = \frac{98}{334\,000} = 0.000\,29\,kg = 0.29\,g.\ \ \frac{0.29}{4.0} \times 100 = \mathbf{7.3\,\%}$ *of the hailstone melts.*

12) **Calculate** the temperature rise of a car's brake discs when the car, of mass $1200\,\text{kg}$, brakes from $20\,\text{m/s}$ to a stop. All of its kinetic energy is transferred to the iron brake discs, of total mass $6.0\,\text{kg}$. c of iron $= 450\,\text{J/kg}\,^\circ\text{C}$. **[5]**

$E_\text{k} = \frac{1}{2}\,m\,v^2 = \frac{1}{2} \times 1200 \times 20^2 = 240\,000\,\textit{J}$. $\Delta\theta = \dfrac{\Delta E}{m\,c} = \dfrac{240\,000}{6.0 \times 450} = \mathbf{89}\ ^\circ\mathbf{C}$ *(to 2 s.f.).*

AQA Trilogy 8464 · aP3-13 to aP3-19 · $E = m\,L$ and $\Delta E = m\,c\,\Delta\theta$ are both given on the equation sheet.

Total _____ / 36